

Virtual Trainer with Real-Time Feedback using Kinect Sensor

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Abstract—With the advancement of technology, everyday life is becoming convenient day by day. However, on the other side, people suffering from various fitness related problems such as obesity, survival and muscular pains due to lack of physical exercise and availability of trainers. In this paper, we propose a virtual trainer with real-time feedback and the assessment score to different exercise postures using Kinect sensor. Recognition of the exercise has been performed using Random Forest (RF) classifier. The classified gestures provide the actual 3D view of the posture on the screen with a matching score such that the user can correct his/her posture by utilizing the feedback. Data of 9 basic exercise postures have been collected from 10 individuals. The performance of the system has been evaluated in user-independent training mode with an average accuracy of 96%. The whole system facilitates individuals an interactive exercise system and a novel contribution to the Human-Computer-Interaction (HCI) research domain.

Keywords: Assistive systems, healthcare, Kinect sensor, Random Forest

I. INTRODUCTION

Nowadays, people are becoming less active physically due to the advancement of convenient technology. Lack of adequate physical exercise leads to unfitness and causes various illness. In everyday routine, exercise plays an important role as it helps people remain in shape, fit and also helps in lower the risk of heart disease, type-II diabetes, stroke, and even a few types of cancers. Therefore health benefits of regular exercise cannot be ignored regardless of age, sex or physical ability [4]. Research findings have revealed that physical activity can also boost self-esteem, mood, sleep quality and energy, as well as reducing the risk of stress, depression, dementia and Alzheimers disease [23]. In order to get fit, people need to go fitness centers and find personal trainers for getting instructions and plans of their own fitness [19]. However, due to unavailability of adequate fitness guides at affordable cost people tends to ignore health benefits.

Currently, available fitness solutions include online videos, mobile applications and books, etc. However, such solutions are not interactive with the user and do not provide any real-time feedback for error correction.

With the advancement of low-cost depth sensing technology such as Microsoft Kinect [8], it is now possible to develop smart healthcare systems with real-time feedback and improved user experiences [6]. The sensor is able to provide 3D point cloud of the observed scene appeared in its field of view. Jin et al. [9] have proposed a virtual trainer for different actions using Kinect sensor. The authors have utilized 20 joints of the 3D skeleton as features. Euclidean distance has been computed to get a matching score and the classification of actions has been performed using Adaboost classifier. In [5], an upper limb rehabilitation system for patients with cerebral palsy has been proposed using Kinect. The system has been designed to measure 3 degrees of freedom (DoF) in upper limb rehabilitation, including elbows and shoulders. The system was able to count correct arm movements. A motor-rehabilitation system for knee osteoarthritis (OA) is proposed in [10]. Lower limbs 3D joints from the skeleton have been extracted as features and Principal Component Analysis (PCA) has been applied to reduce the dimensionality of the data. A study has been performed on 5 subjects, where an accuracy of 95.6% has been recorded using k-NN classifier with k=2.

One of the primary goals of a smart assistive system is to provide a real-time feedback to the users. It helps the users to correct errors and perform all exercises in an appropriate manner. In [1], the authors have proposed rehabilitation system for disabled children which complement the role of a therapist. Kinect sensor has been used in their system which allows a physiotherapist to record the rehabilitation exercises which are then transmitted over the network to a virtual rehabilitation center. Later, trainees are allowed to visit the virtual center where exercise has been performed by an avatar on the screen. The system is able to record kids sessions and send the data to the therapist for evaluation purpose. Similarly, a Sign Language Recognition (SLR) has been proposed to communicate hearing impaired people with normal people using Kinect [12], [13], [18] and Leap motion sensor [2], [14]. The authors have extracted fingertips as features and the classification of hand gestures have been performed using Hidden Markov Model. In [21], the authors have proposed a activity recognition system for weight lifting exercise using

Kinect sensor. In addition to activity recognition, the system is able to detect mistakes and provides feedback to the user. Five classes of boxing exercise have been considered in their work with an average accuracy of 78.2% that has been recorded using Random Forest classifier.

3D air-writing segmentation and recognition frameworks to assist hearing impaired people have been proposed in [15], [16]. The authors have used Leap motion sensor to record the data and applied a window based segmentation technique to break the sentences into words. Later, the words are recognized using Hidden Markov Model (HMM) classifier. In [20], the authors have proposed a palm rehabilitation technique using Leap motion sensor. The recognition of gestures has been performed using HMM classifier. To assist differently abled persons, brain wave based predictive analysis of advertisement videos and marketing products has also been proposed in literature using Electroencephalogram (EEG) signals [7], [11], [17], [24]. The authors have recorded the EEG signals while the participants watching the products and videos. The analysis of the recorded EEG data has been performed using Random Forest classifier.

In this paper, we present a virtual trainer for doing in-house exercise using Kinect sensor. The system is able to provide real time feedback in two forms. First, it provides the 3D view of the best matched posture from the repository that acts as a visual assistance for the user. Secondly, it presents the matching score of the test posture such that the user can assess his/her exercise posture in terms of performance accuracy. The main contributions of the paper are as follows:

- 1) Firstly, we present a interactive assistive system for daily exercise that is able to provide real time feedback for error correction and self assessment.
- 2) Secondly, a repository of actual exercise postures has been prepared. Next, Random Forest (RF) classifier has been trained to evaluate the performance in user-independent training approach. The confidence score for each test exercise has also been extracted from RF classifier for the self assessment of exercise by the user.

Rest of the paper is organized as follows. We present the details of the proposed system along with features and description of the classification scheme in Section II. Results are discussed in Section III. Finally, we conclude in Section V by discussing future possibilities of the presented work.

II. PROPOSED SYSTEM

In this section, we present the details of the proposed virtual trainer system using Kinect sensor. The flow diagram of the system is depicted in Fig. 1, where the training data has been prepared by capturing the actual exercise postures from experts using Kinect sensor. Next, 3D skeleton of the human posture that consists of twenty 3D points has been extracted using the sensor's Application Programming Interface (API) which is then preprocessed and relevant features are extracted. To evaluate the system performance, the testing postures of the users have been captured and classified with the help of RF classifier. Finally, the classified gestures have been displayed in

3D view such that the users can correct the exercise postures. A score confidence of each exercise posture has also been displayed that denotes the correctness of the exercise being performed by a user.

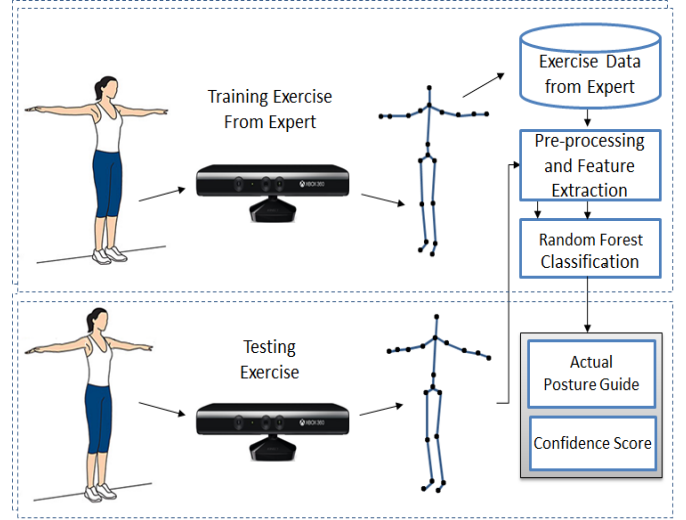


Fig. 1. Flow diagram of the proposed virtual trainer using Kinect.

A. Preprocessing and Feature Extraction

In this section, we present the details of the preprocessing and feature extraction techniques that have been applied on the 3D skeleton before classification. The details are as follows.

B. Skeleton Translation

Since all users are free to exercise at any position of the device viewing field in the XZ plane, it creates confusion during the classification process. Therefore, we have transformed the coordinates of the 3D skeleton from the sensors frame of reference to a new frame of reference fixed to the signer. It has been performed by translating the 3D spine point (C) of the skeleton to the center and shifting the rest of data points with respect to this new origin. Using this, we have translated all skeleton joints to a common origin.

1) *Joint Selection*: All extracted points of the 3D skeleton may not be informative due to various noises present in the device and to get the structure of the human body. Also some points are close to each others, thus contain redundant information. Therefore, we have skipped a few joint positions and kept only 12 joints using an evolutionary algorithm. The selected joints ($J_k, k = 1, 2, \dots, 12$) with their positions in their skeleton are depicted in Fig. 2(a).

2) *Angular Features*: From the selected joints, 8 angular features ($A_i, i = 1, 2, \dots, 8$) have been extracted by calculating the cosine angle between two vectors using (1), where $\vec{X}$ and $\vec{Y}$ are the two vectors formed using three joints. The pictorial representation of all the extracted angles are depicted in Fig. 2 (b).

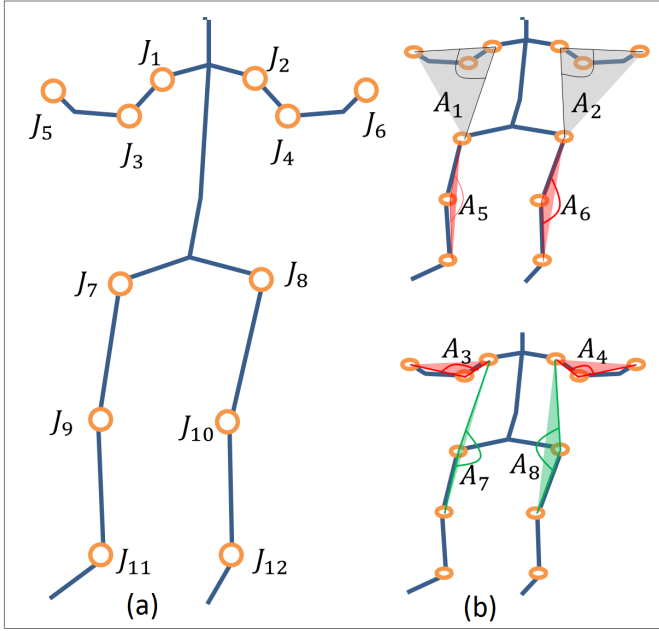


Fig. 2. Feature computation: (a) selected joints position (b) description of the angular features.

$$A_i = \cos \alpha = \frac{\vec{X} \cdot \vec{Y}}{|\vec{X}| |\vec{Y}|} \quad (1)$$

The final feature set (F_T) is composed of the 3D selected joints (J_k , $k = 1, 2, \dots, 12$) and the 8 angular features (A_i , $i = 1, 2, \dots, 8$) as defined in (2). Thus, F_T consists of 44 dimensions.

$$F_T = \{J_k, A_i\} \quad (2)$$

C. Random Forest (RF) Guided Exercise Classification

RF classifier is defined as an ensemble of classification trees. Each tree contributes in final decision of assigning class to input data [3]. It uses a random subset of input features for the division of every node. Additionally, it uses bagging technique for growing trees using random training subsets by random resampling of the original dataset with replacement [22]. The RF classifier uses Gini Index (GI) for split selection that measures the impurity of a data sample with respect to the classes. Given a node t , the GI can be computed using (3). Using this, the test samples are passed down to the leaf node and the classification is performed with the help posterior probabilities.

$$GI(t) = 1 - \sum_{x=1}^Q p^2(x|t) \quad (3)$$

In (3), $p(x|t)$, $\{x = 1, 2, \dots, Q\}$ denotes the estimated class probabilities for Q number of classes.

III. RESULTS

In this section, we present the details of the dataset that has been evaluated using the proposed framework. Next, analysis of results has been performed in user-independent training approach using RF classifier. By user-independent training, we mean that the exercise data of a user is not included in training while performing testing for the same user.

A. Dataset Description

In this work, we have selected 9 basic exercises that should ideally be performed by everyone in daily life. We have enrolled 10 volunteers in the dataset preparation. Out of these 10 participants, 4 users are experts in doing these exercises that has helped us in making the training data. Rest of the participants are student volunteers and we have used their data to evaluate the performance of the system. We have recorded 10 instances of each exercise from every participant. Therefore, a total of 900 (i.e. $10 \times 9 \times 10$) samples have been recorded. A pictorial representation of the exercises involved in the dataset are depicted in Fig. 3. Variations in the exercise postures during data collection can be seen in the Fig. 4, where three users are performing the same exercise, however, with different postures.

IV. EXERCISE CLASSIFICATION USING RF

Exercise classification using RF has been performed using the feature vector F_T by varying the number of trees in the forest from 1 to 50. The classifier has been trained on the data of 4 expert users who have performed these exercises as per the guidelines. The accuracy of the systems has been computed in user-independent mode. Therefore, the classification of exercise has been performed for each user. User-wise accuracies are depicted in Fig. 5, where an average accuracy of 96% has been recorded across all users. It can be verified from the Fig. 5 that the minimum accuracy of 80% has been recorded for user U1, whereas, an accuracy of 100% has been recorded for user U6. It is because, user U1 has not performed all gestures in correct manner, which has created confusion during the classification.

In order to assess the performed exercise, we have computed the confidence score from RF classifier for each test example. An average confidence score for each user has been evaluated and depicted in Fig. 5, where the maximum score of 90% has been recorded for user U3. This information can help a user to correct his/her error and it may help in keeping daily progress. In our system, the actual posture of the classified exercise is displayed on the user screen along with the confidence score as depicted in Fig. 6, where two users are performing the same exercise. However, the scores are different because of the user in Fig. 6(a) is not performing the exercise in correct manner. Hence, it has a low confidence score in comparison to the user depicted in Fig. 6(b).

In addition, we have shown the effectiveness of the featured data on the accuracy of the system. The classification rates have been computed on raw as well as featured data. The results are depicted in Fig. 7, where a gain of 12% has been



Fig. 3. Pictorial representation of the exercise involved in the dataset.

recorded using our model when compared to the classification result on raw data.

A comparison with other state of the art works has been performed. Velloso et al. [21] have proposed a activity recognition system for weight lifting exercises using Kinect. The author have recorded the data from 6 users while performing 5 different exercise. The comparison is presented in Table I, where the proposed scheme performs better with more users and exercise.

TABLE I
COMPARISON WITH OTHER METHOD(S)

Authors	Method	Number of Subjects	Number of Activities	Accuracy (%)
Velloso et al. [21] 2013	Eular Angles, RF	6	5	78.20
Proposed work	Joint Angles, RF	10	9	96.00

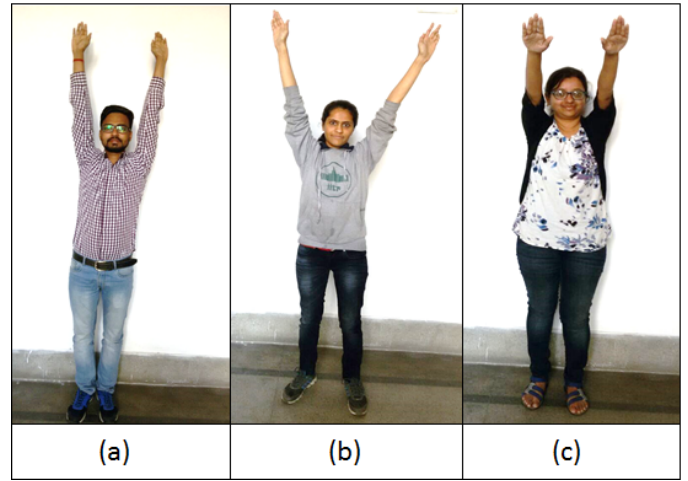


Fig. 4. Variations in the exercise postures while performing same exercise by three different users.

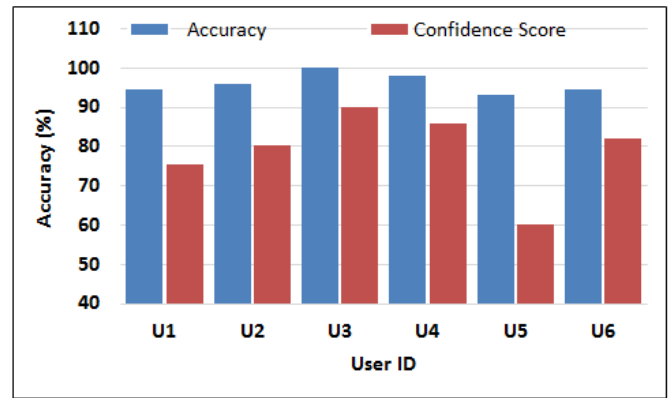


Fig. 5. Average accuracy of all users along with their confidence scores.

V. CONCLUSION

Due to significant technological advancement, people tend to ignore health benefits of physical exercise. Even people are avoiding simple exercises like running, walking, body stretching etc. Moreover, due to busy life, people do not find adequate time to attend gym and personal trainers to correct them while doing exercise. In this paper, we have proposed a virtual trainer with real-time feedback of the posture and the confidence score. The feedback posture helps a user to understand the actual exercise, whereas the confidence displays the correctness of the exercise such that the user can correct his/her errors during exercise. The methodology has been tested on 6 participants in user independent training with an average accuracy of 96% using Random Forest classifier. In future, the system can be extended to provide real-time feedback for more complex exercises with animated avatars that can provide vocal training to the participants by analyzing their postures during exercise.

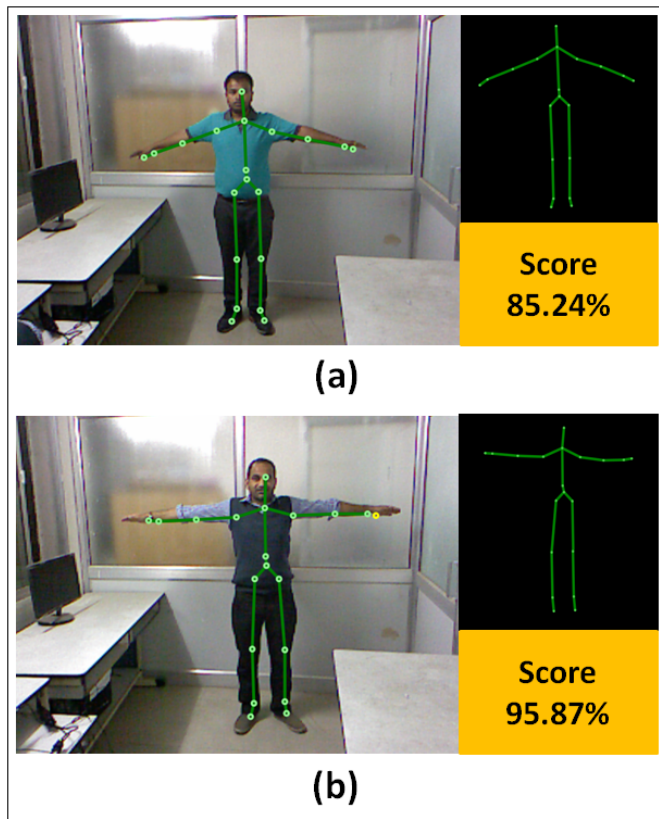


Fig. 6. Examples of the complete system of virtual trainer with feedback and score: (a) novice user (b) expert user.

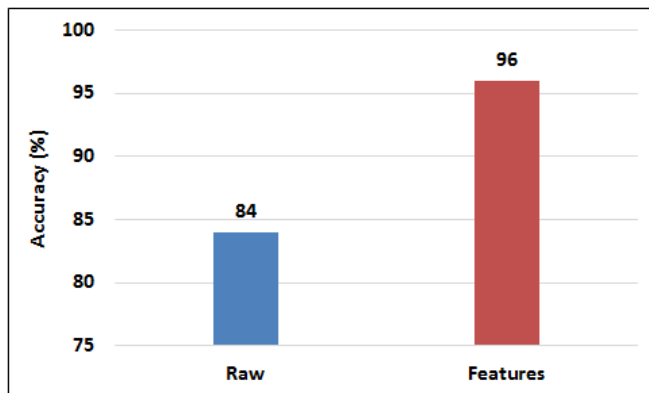


Fig. 7. Comparative performance analysis between raw and featured data.

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